

2007

**HIGHER SCHOOL CERTIFICATE
EXAMINATION**

Biology

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black or blue pen
- Draw diagrams using pencil
- Board-approved calculators may be used
- Write your Centre Number and Student Number at the top of pages 9, 13, 17 and 19

Total marks – 100

Section I Pages 2–20

75 marks

This section has two parts, Part A and Part B

Part A – 15 marks

- Attempt Questions 1–15
- Allow about 30 minutes for this part

Part B – 60 marks

- Attempt Questions 16–27
- Allow about 1 hour and 45 minutes for this part

Section II Pages 21–31

25 marks

- Attempt ONE question from Questions 28–32
- Allow about 45 minutes for this section

Section I
75 marks

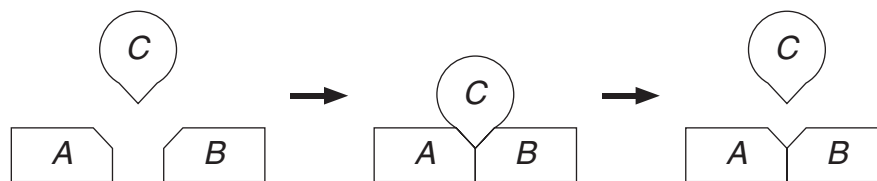
Part A – 15 marks

Attempt Questions 1–15

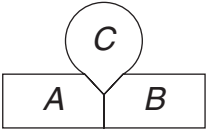
Allow about 30 minutes for this part

Use the multiple-choice answer sheet for Questions 1–15.

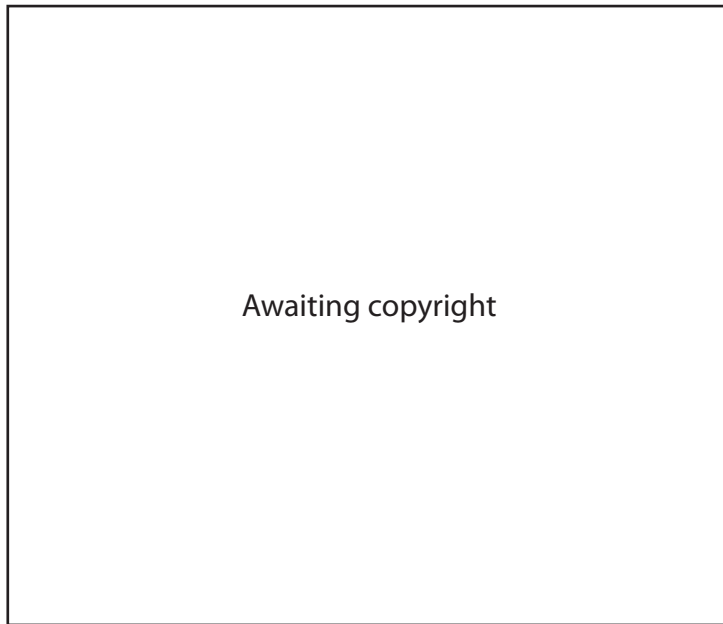
- 1** The diagram shows one example of enzyme action as demonstrated by the ‘Lock and Key’ model.



Which part of the diagram represents the substrate?

- (A) 
- (B) 
- (C) 
- (D) 
- 2** Why do cells contain many different enzymes?
- (A) Enzymes are temperature specific.
- (B) Enzymes are specific in their action.
- (C) Enzymes are sensitive to pH changes.
- (D) Enzymes are sensitive to substrate concentration.

- 3 This is a longitudinal section of plant stem ($\times 200$).



What is the name and function of the tissue labelled *W*?

	<i>Name</i>	<i>Function</i>
(A)	Xylem	Transports water and mineral ions
(B)	Phloem	Transports water and mineral ions
(C)	Xylem	Transports simple sugars
(D)	Phloem	Transports simple sugars

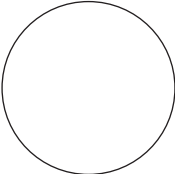
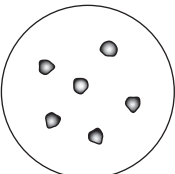
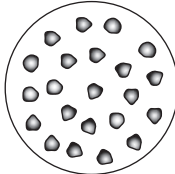
- 4 The Wollemi Pine (*Wollemia nobilis*) is easily killed by the fungus *Phytophthora* which lives in the soil. The last small population of Wollemi Pines grows in a remote part of a national park in NSW. Scientists studying this natural population use strategies to prevent the trees becoming infected with *Phytophthora*.

Which procedure would be most effective in preventing the spread of this fungus to the Wollemi Pines?

- (A) Inspecting soil samples in the area
- (B) Commercially producing and distributing the Wollemi Pine
- (C) Washing soil from scientists' shoes before they walk in the area
- (D) Preventing the importation of infected Wollemi Pines into Australia

- 5 Students performed an investigation to compare the effectiveness of two water treatments for purifying pond water.

Three samples of pond water, *A*, *B* and *C*, were collected and each used to inoculate an agar plate. The plates were incubated at 25°C and examined three days later. The number of visible bacterial colonies on each plate was counted and the results tabulated.

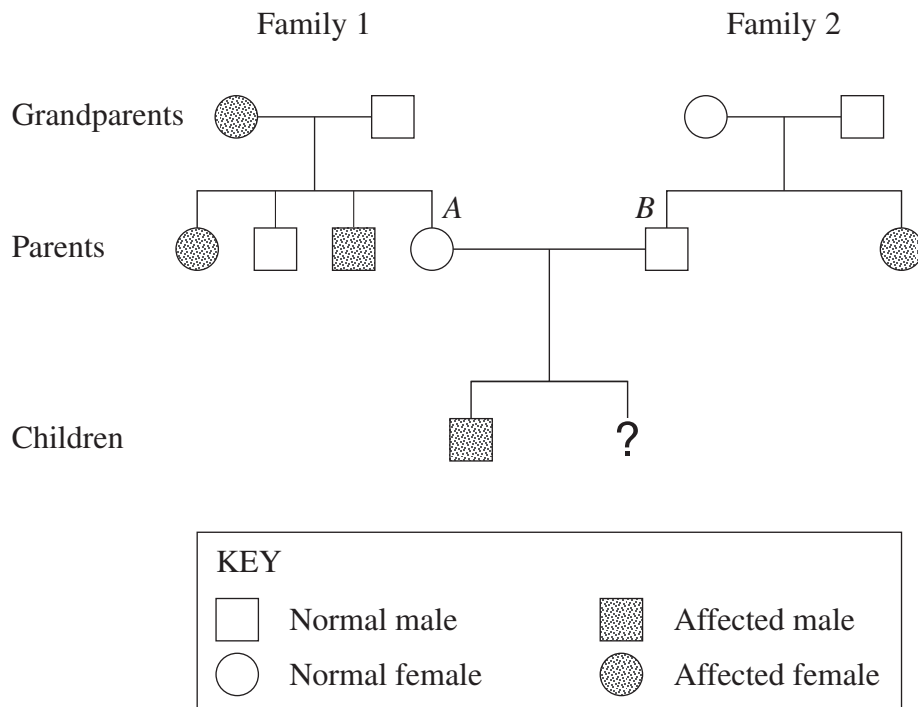
<i>Sample</i>	<i>A</i>	<i>B</i>	<i>C</i>
<i>Treatment</i>	5 grams of pool chlorine per litre of water	Boiling for one minute	No treatment
<i>Number of visible bacterial colonies</i>	0 	6 	22 

What is the dependent variable in this investigation?

- (A) The use of a control sample
 - (B) The number of visible bacterial colonies
 - (C) The use of sterile agar plates for each sample
 - (D) Treating the water by boiling or adding pool chlorine
- 6 Which biological term is best described by ‘engulfing and destruction of bacteria or other foreign bodies’?
- (A) Vaccination
 - (B) Phagocytosis
 - (C) Antibody production
 - (D) An inflammation response

- 7 Which leaf structures are adaptations to assist in the conservation of water?
- (A) Central vein, irregular leaf shape
 - (B) Large air spaces, pointed leaf tip
 - (C) Spongy mesophyll, vascular bundle
 - (D) Sunken stomates, thick waxy cuticle
- 8 Which observations can be used to demonstrate Koch's contribution to understanding the cause of disease?
- (A) Polio vaccinations trigger an immune response.
 - (B) Some mosquitoes carry a pathogen that is often fatal to people.
 - (C) A lack of vitamin C is found in all people suffering the nutritional disease scurvy.
 - (D) The bacteria, *Helicobacter pylori*, is present in the stomach of all people diagnosed with stomach ulcers.
- 9 Current reproductive techniques can be used to alter the genetic composition of a population. Some of these methods were also used in the nineteenth century by Gregor Mendel.
- How did Mendel use reproductive techniques in his experiments?
- (A) He artificially inseminated the pea plants to achieve wrinkled seeds.
 - (B) He cloned the pea plants with round seeds to increase their food supply.
 - (C) He created a transgenic species by mixing tall pea plants and short pea plants.
 - (D) He artificially pollinated the pea plants to test for different genotypes in the offspring.
- 10 Which statement best describes the relationship between proteins and polypeptides?
- (A) Proteins are composed of polypeptides.
 - (B) Polypeptides are composed of proteins.
 - (C) Proteins, unlike polypeptides, are composed of amino acids.
 - (D) Polypeptides, unlike proteins, are composed of amino acids.

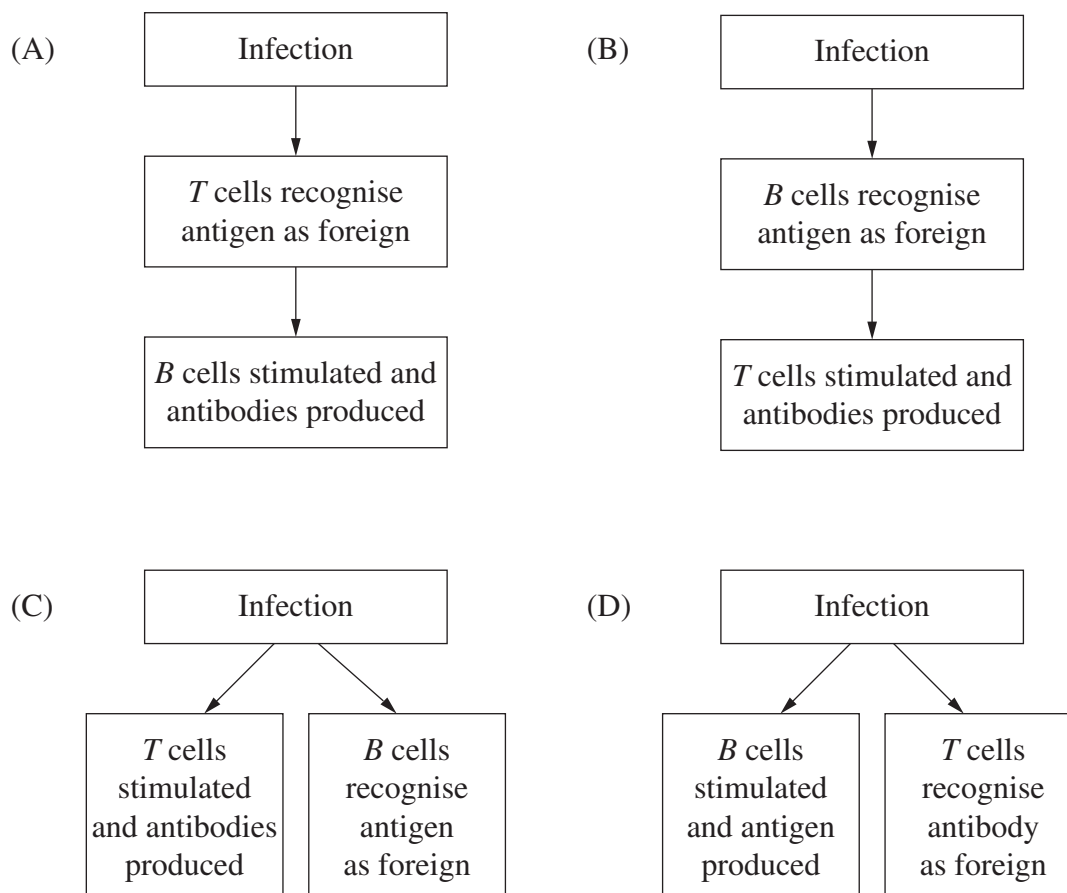
- 11 The family tree shows the inheritance of a genetic characteristic.



What is the probability that a daughter of parents *A* and *B* would be affected?

- (A) 0%
(B) 25%
(C) 50%
(D) 100%

- 12 Which flowchart correctly shows an interaction between *B* and *T* lymphocytes during an immune response?



- 13 The effectiveness of a new insecticide was tested on a large population of mosquitoes over a number of breeding cycles. At first the population of mosquitoes was reduced dramatically by the use of the insecticide. After a number of breeding cycles the population then began to increase until the insecticide appeared to have little effect.

How would the Darwin/Wallace theory of evolution by natural selection explain these observations?

- (A) Some of the original population were isolated from the insecticide as a control group.
- (B) Some of the original population had already reproduced before the insecticide was used.
- (C) Some of the original population were resistant to the insecticide and passed this on to their offspring.
- (D) Some of the original population adapted to the insecticide and survived to produce offspring.

- 14** At the end of a marathon race a runner's body is dehydrated.

How does the body control the two hormones, ADH and aldosterone, to help to re-establish normal water balance?

- (A) ADH is released and aldosterone is inhibited.
- (B) ADH is inhibited and aldosterone is released.
- (C) Both ADH and aldosterone are released.
- (D) Both ADH and aldosterone are inhibited.

- 15** How have Walter Sutton and Theodor Boveri contributed to the understanding of inheritance?

- (A) By determining the structure of DNA
- (B) By improving knowledge of sex linkage
- (C) By demonstrating the effect of environment on phenotype
- (D) By identifying the importance of chromosomes in inheritance

Biology

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Centre Number

Section I (continued)

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Part B – 60 marks

Student Number

Attempt Questions 16–27**Allow about 1 hour and 45 minutes for this part**

Answer the questions in the spaces provided.

Marks**Question 16 (3 marks)**

A student working in a restaurant kitchen is required to wear disposable gloves and hat when preparing food.

- (a) Explain how this practice assists in the control of disease.

2

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- (b) Identify another hygiene practice that reduces the risk of infection.

1

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Question 17 (3 marks)

Construct a table to distinguish between the environmental conditions experienced by marine and fresh water fish, and the resulting urine production of each fish. **3**

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Question 18 (4 marks)

(a) Name a blood product extracted from donated blood and outline how it could be used to restore normal body function. **2**

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(b) Propose TWO reasons why research is needed to develop alternatives to donated blood. **2**

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Question 19 (6 marks)

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|-----|--|----------|
| (a) | Name ONE example of a disease caused by a macro-parasite. | 1 |
| | | |
| (b) | List TWO features of prions that distinguish them from protozoans. | 2 |
| | | |
| | | |
| (c) | Most pathogens must first be transmitted to and enter the human body before they trigger an immune response. | 3 |

Relate this statement to a named infectious disease you have studied.

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Centre Number

Section I – Part B (continued)

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Student Number

Question 20 (6 marks)

Please turn over

Question 20 (6 marks)

- (a) The table shows that 20% of the bases in a section of double-stranded DNA are adenine (A). **3**

Complete the table below by identifying the other three base types and calculating the percentage of each base type in the section of double-stranded DNA.

<i>Bases</i>	<i>Percentage (%)</i>
A	20

- (b) Construct a simple flowchart to describe the process of DNA replication. **3**

Question 21 (6 marks)



- (a) Which disease in the table is the most dangerous? Give TWO reasons for your answer. **2**

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- (b) Evaluate the effectiveness of a vaccination program for ONE named disease from the table. **4**

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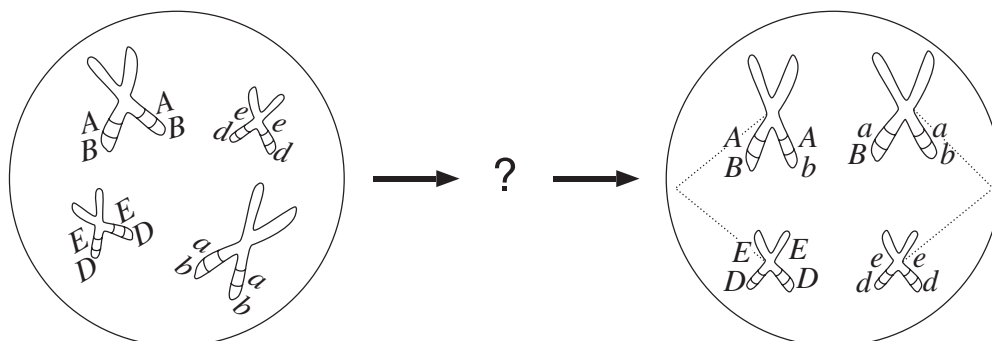
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Question 22 (8 marks)

The diagram shows two steps of the process of meiosis occurring in a cell with four chromosomes.



- (a) Describe the behaviour of the chromosomes between the steps shown. 2

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- (b) List FOUR possible combinations of alleles that would be found in the gametes resulting from this process. 2

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- (c) Explain ONE advantage of the process of meiosis to the species. 2

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- (d) Distinguish between the terms *allele* and *gene*. 2

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Centre Number

Section I – Part B (continued)

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Student Number

Marks

Question 23 (4 marks)

Nothofagus gunnii is a deciduous beech tree that grows in Tasmania. When environmental temperatures decrease at the beginning of winter the beech tree drops its leaves.

4

Design a procedure, using potted beech seedlings, to investigate the temperature at which leaves begin to drop.

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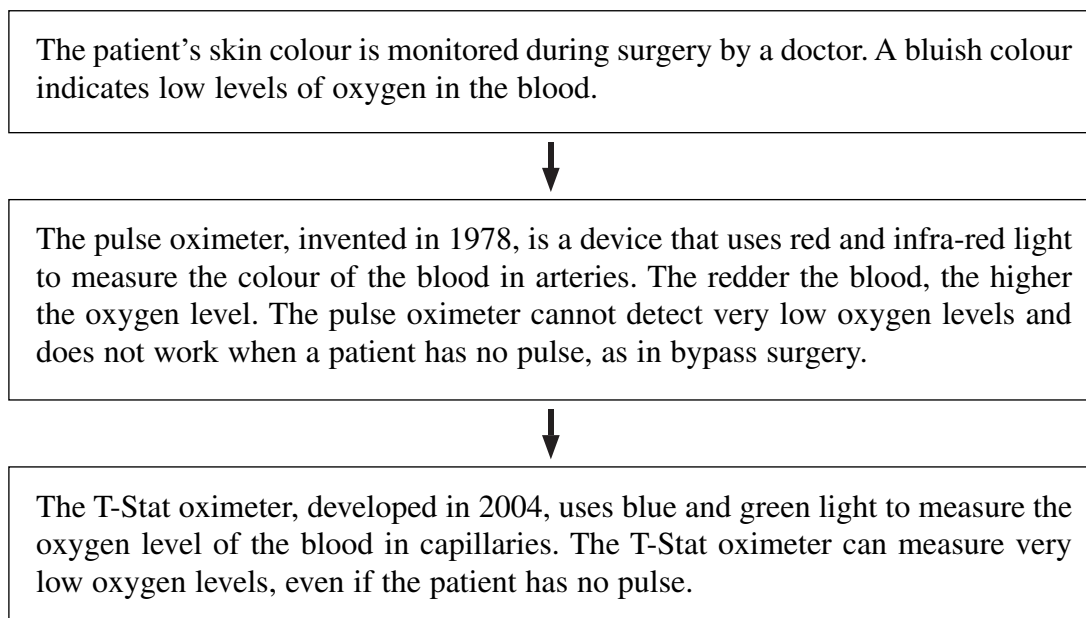
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Question 24 (6 marks)

The flowchart shows the development of technology used to measure oxygen concentration in blood during surgery.



- (a) Why is it important to monitor oxygen levels in the blood during surgery? 1

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- (b) Explain ONE advantage of the T-Stat oximeter over the pulse oximeter. 2

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- (c) Explain TWO changes in the chemical composition of blood as it moves along a capillary. 3

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Centre Number

Section I – Part B (continued)

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Student Number

Marks**Question 25** (3 marks)

Epidemiological studies indicate that there is a relationship between smoking and the incidence of lung cancer.

3

What information would have been gathered to establish this relationship?

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Question 26 (3 marks)

Describe how genes assist in the maintenance of health.

3

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8

This image shows a full page of white paper with horizontal dotted lines, typical of primary school writing paper. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

Biology

Section II

25 marks

Attempt ONE question from Questions 28–32

Allow about 45 minutes for this section

Answer the question in a writing booklet. Extra writing booklets are available.

	Pages
Question 28 Communication	22–23
Question 29 Biotechnology	24–25
Question 30 Genetics: The Code Broken?	26–27
Question 31 The Human Story	28–29
Question 32 Biochemistry	30–31

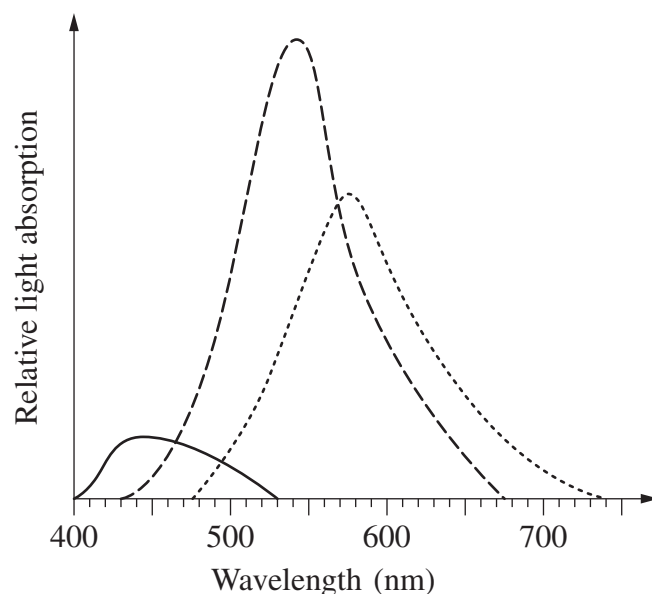
Question 28 — Communication (25 marks)

- | | | |
|------|--|----------|
| (a) | (i) Name TWO refractive media in the human eye. | 2 |
| | (ii) Explain how a problem with ONE of these media may contribute to poor eyesight or blindness. | 3 |
|
 | | |
| (b) | (i) Describe an investigation you conducted to distinguish parts of the brain and locate the regions involved in speech, sight and sound perception. | 3 |
| | (ii) How would you evaluate the relevance and reliability of the information gathered in this investigation? | 3 |
|
 | | |
| (c) | Assess how structures in the human body that produce, detect and perceive sound enable effective communication. Include examples of these structures in your answer. | 7 |

Question 28 continues on page 23

Question 28 (continued)

- (d) The graph shows the relative light absorption by cones in the human eye.



KEY

- Blue pigment cone
 ---- Green pigment cone
 Red pigment cone

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- | | | |
|-------|---|----------|
| (i) | Estimate the wavelength at which the green pigment cone absorbs most light. | 1 |
| (ii) | Describe the pattern of light absorption by the red pigment cone, and suggest why all three pigment cones are necessary for colour vision. | 2 |
| (iii) | Explain the relationship between the occurrence of colour vision in animals and their use of colour for communication. Include examples in your answer. | 4 |

End of Question 28

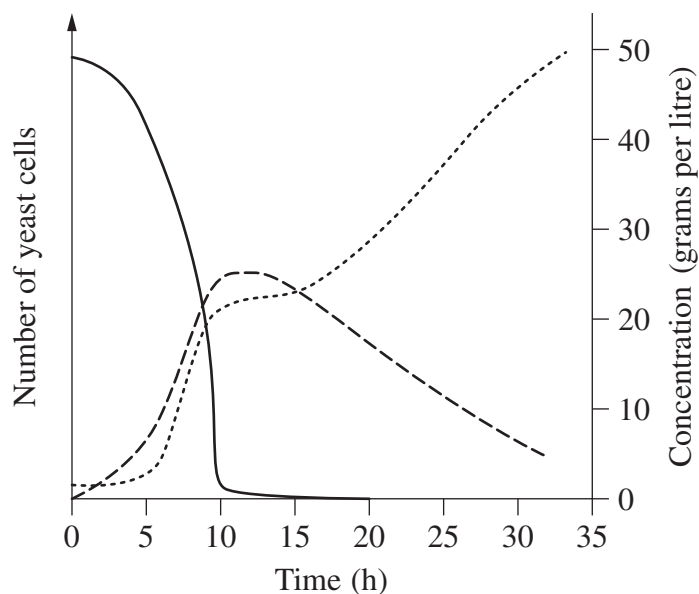
Question 29 — Biotechnology (25 marks)

- | | | | |
|------|------|--|----------|
| (a) | (i) | Name TWO organic compounds produced by fermentation techniques developed since the early eighteenth century. | 2 |
| | (ii) | Explain how the use of ONE of these organic compounds had an impact on society at the time of its introduction. | 3 |
|
 | | | |
| (b) | (i) | Describe how you gathered and analysed information to outline the purpose of a current application of transgenic technology. | 3 |
| | (ii) | How would you evaluate the relevance and reliability of the information gathered in this investigation? | 3 |
|
 | | | |
| (c) | | Assess the biotechnological processes used in aquaculture. Include examples in your answer. | 7 |

Question 29 continues on page 25

Question 29 (continued)

(d) The graph shows data recorded during fermentation by yeast cells.



KEY

..... Number of yeast cells

———— Glucose

----- Ethanol

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- (i) Estimate the time at which ethanol concentration is at its maximum. **1**
- (ii) Describe the trend in glucose concentration, and suggest a reason for this change. **2**
- (iii) Explain how changes in technology have modified traditional uses of biotechnology. **4**

End of Question 29

Question 30 — Genetics: The Code Broken? (25 marks)

- | | | | |
|-----|------|---|----------|
| (a) | (i) | Name TWO examples of characteristics determined by multiple alleles in an organism other than humans. | 2 |
| | (ii) | Explain how ONE of these characteristics provides variability in phenotypes. | 3 |
| (b) | (i) | Describe how you gathered and analysed information to describe the processes used in the cloning of an animal. | 3 |
| | (ii) | How would you evaluate the relevance and reliability of the information gathered in this investigation? | 3 |
| (c) | | Assess how the mapping of the Human Genome and gene therapy assist in managing a genetic disease OR a form of cancer OR AIDS. | 7 |

Question 30 continues on page 27

Question 30 (continued)

- (d) The graph shows maternal age and incidence of Down syndrome (Trisomy 21) births.



- | | |
|---|----------|
| (i) Estimate the number of infants with Down syndrome per 1000 births for 40-year-old mothers. | 1 |
| (ii) Describe the trend in the number of Down syndrome births, and define the term <i>trisomy</i> . | 2 |
| (iii) Discuss the impact on the genome of transposable genetic elements. | 4 |

End of Question 30

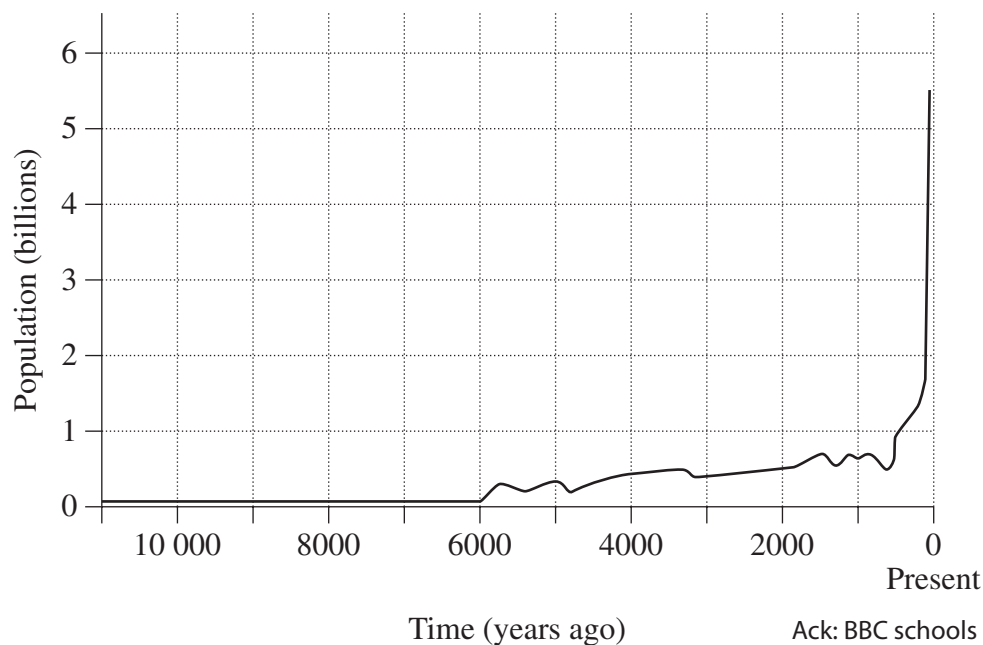
Question 31 — The Human Story (25 marks)

- | | | |
|-----|---|----------|
| (a) | (i) Name TWO hominids known from fossil evidence. | 2 |
| | (ii) Describe how ONE of these hominids differs from <i>Homo sapiens</i> . | 3 |
| (b) | (i) Describe how you processed information and used available evidence to assess the contribution of ONE scientist to our increased understanding of human evolution. | 3 |
| | (ii) How would you evaluate the relevance and reliability of the information gathered in this investigation? | 3 |
| (c) | Assess the impact of cultural development throughout human evolution. Include examples in your answer. | 7 |

Question 31 continues on page 29

Question 31 (continued)

- (d) The graph shows estimated human population for the last 10 000 years.



- (i) Estimate the human population 1000 years ago. **1**
- (ii) Describe a change in human population numbers over the last 10 000 years, and suggest ONE reason for this change. **2**
- (iii) Discuss the potential impact of a modern technology on future human populations. **4**

End of Question 31

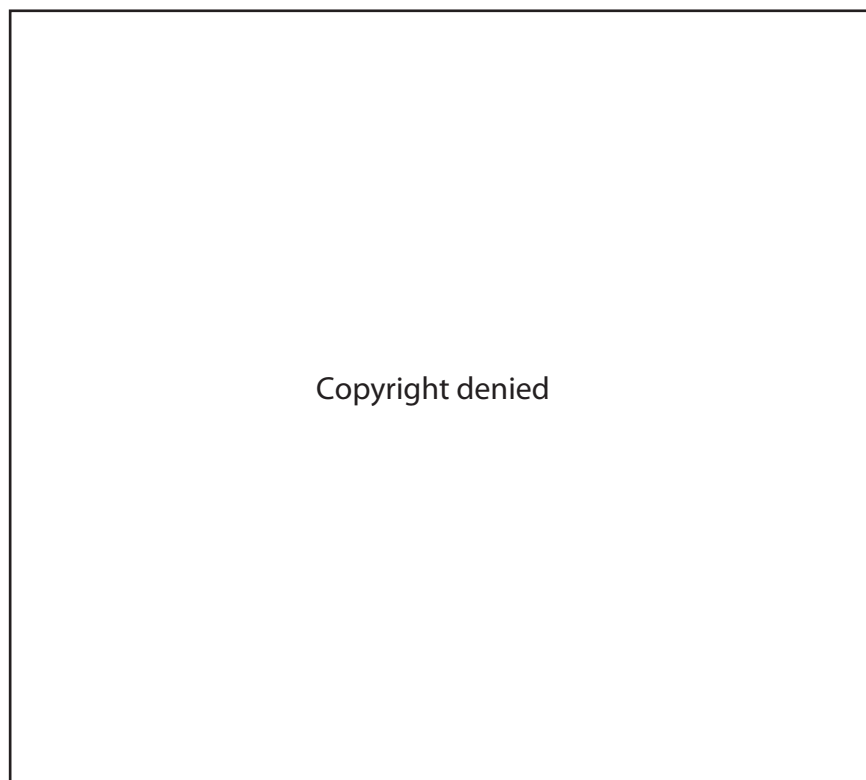
Question 32 — Biochemistry (25 marks)

- | | | | |
|-----|------|---|----------|
| (a) | (i) | Name TWO products of photosynthesis. | 2 |
| | (ii) | Explain how ONE of these products is used in living organisms. | 3 |
| (b) | (i) | Describe how you gathered and presented information to compare the size, shape and distribution of chloroplasts in different angiosperms. | 3 |
| | (ii) | How would you evaluate the relevance and reliability of the information gathered in this investigation? | 3 |
| (c) | | Assess the role of isotopes in developing an understanding of photosynthesis. Include examples in your answer. | 7 |

Question 32 continues on page 31

Question 32 (continued)

- (d) The graph shows absorption and action spectra for photosynthesis.



- | | | |
|-------|---|----------|
| (i) | Estimate the wavelength for the maximum rate of photosynthesis. | 1 |
| (ii) | Describe the absorption of chlorophyll <i>a</i> over the spectrum of visible light, and identify the location within a chloroplast where this occurs. | 2 |
| (iii) | Discuss the role of pigments, other than chlorophyll <i>a</i> , in photosynthesis. | 4 |

End of paper

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**2007 HSC Notes from
the Marking Centre
Biology**

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2007 HSC NOTES FROM THE MARKING CENTRE

BIOLOGY

Introduction

This document has been produced for the teachers and candidates of the Stage 6 course in Biology. It contains comments on candidate responses to the 2007 Higher School Certificate examination, indicating the quality of the responses and highlighting their relative strengths and weaknesses.

This document should be read along with the relevant syllabus, the 2007 Higher School Certificate examination, the marking guidelines and other support documents which have been developed by the Board of Studies to assist in the teaching and learning of Biology.

General Comments

In 2007, approximately 14,440 candidates attempted the Biology examination. The most popular Options were Communication (60%), and Genetics: The Code Broken? (20%).

Teachers and candidates should be aware that examiners may write questions that address the syllabus outcomes in a manner that requires candidates to respond by integrating their knowledge, understanding and skills developed through studying the course, including the Prescribed Focus Areas. This reflects the fact that the knowledge, understanding and skills developed through the study of discrete sections should accumulate to a more comprehensive understanding than may be described in each section separately. It is important to understand that the Preliminary HSC course is assumed knowledge for the HSC course.

Teachers and candidates are reminded that mandatory skills content in Module 9.1 is examinable in both the Core and Option questions.

Candidates need to be reminded that the answer space provided and the marks allocated are guides to the maximum length of response required. Candidates should use examination time to analyse the question and plan responses carefully, working within that framework to produce a clear and concise response. This may include the use of dot points, diagrams and/or tables, and avoid internal contradictions. This is particularly important in holistic questions that need to be logical and well structured.

Better responses indicate that candidates are following the instructions provided on the examination paper. In these responses, candidates:

- do not repeat the question as part of the response
- look at the structure of the whole question and note that in some questions the parts follow from each other (ie responses in part (a) lead to the required response in part (b) etc).

In Section II, the Option question is divided into a number of parts. Candidates should clearly label each part of the question when writing in their answer booklets. In part (c) of the 2007 Option questions, the best responses presented ideas coherently and included the correct use of scientific principles and ideas. Many candidates wrote a lot of information that was not relevant to the question. Some responses showed evidence of rote learning of an anticipated answer based on a single source. These responses did not address the syllabus content and/or outcomes being assessed and hence did not score full marks. Candidates are required to attempt one question only in Section II, but some candidates responded to more than one Option question. Candidates are strongly advised to answer the Option they have studied in class.

Section I – Core

Part A – Multiple-choice questions

Question	Correct Response
1	D
2	B
3	A
4	C
5	B
6	B
7	D
8	D

Question	Correct Response
9	D
10	A
11	B
12	A
13	C
14	C
15	D

Part B

Specific Comments

Question 16

- (a) In the better responses, candidates explained how the practice of wearing gloves assists in the control of disease. In the weaker responses, candidates were unable to identify human skin or hair as a source of pathogens.

Question 17

The better responses constructed a table with appropriate headings which was used to distinguish between environmental conditions and the urine production of each type of fish. The weaker responses did not correctly construct a table.

Question 18

- (a) Better responses named a blood product and outlined its use in restoring normal body function.
- (b) Better responses proposed two valid reasons for research into alternatives to donated blood.

Question 19

- (a) In weaker responses, candidates either named a disease caused by a micro-organism, or named a macroparasite without naming the disease it causes.
- (b) In better responses, candidates listed two features of prions that distinguished them from protozoans. Weaker responses showed a lack of specific knowledge about prions.
- (c) In better responses, candidates related transmission and entry to a specific infectious disease and the appropriate immune response. In weaker responses, transmission and entry to the body were treated in more general terms, and the triggering of an immune response was not addressed or related to a specific disease.

Question 20

- (a) In better responses, candidates showed a good understanding of the proportions of bases in DNA.
- (b) Better responses demonstrated an understanding of DNA structure. In weaker responses, candidates confused DNA replication with protein synthesis, or did not present their information in the form of a flow chart.

Question 21

- (b) In the better responses, candidates evaluated the effectiveness of a vaccination program. In the weaker responses, candidates described or outlined a program without an evaluation.

Question 22

- (a) In better responses, candidates described crossing-over and pairing of homologous chromosomes. In weaker responses, candidates identified these behaviours without a description.
- (b) In better responses, candidates correctly listed the four combinations of alleles possible. In weaker responses, candidates referred to only one pair of chromosomes or were not able to list any of the combinations correctly.
- (c) Better responses recognised variation as an advantage of meiosis.
- (d) In better responses, candidates distinguished between the two terms. In weaker responses, candidates only provided an outline of either an allele or gene, or could not provide a difference between them.

Question 23

Better responses designed a reliable and valid procedure. In weaker responses, candidates did not indicate how the temperature would be changed or did not provide a list of equipment to be used.

Question 24

- (b) Better responses explained the comparative advantage of the T-Stat oximeter. In weaker responses, candidates did not relate the improvement in the T-Stat oximeter to the advantage.
- (c) In better responses, candidates related changes in the chemical composition of the blood to the reasons for the changes. The weaker responses were not clear about the features of the chemical changes.

Question 25

In better responses, candidates identified characteristics of a sound epidemiological study and linked this to the need to establish a relationship between smoking and lung cancer. In weaker responses, candidates had difficulty expressing the aspects of a sound study or simply recited the words in the question.

Question 26

In better responses, candidates used specific examples to describe a clear link between genes and the maintenance of health.

Question 27

In better responses, candidates explained the consequences of temperature change using correct biological terminology for specific examples of both animals and plants, and demonstrated a sound understanding of natural selection.

In weaker responses, candidates simply stated some generalised animal and/or plant responses. Weaker responses incorrectly described the process of organisms adapting to climate change in a Lamarckian way. Some responses described responses to water loss in terms of the structural features of plants, without linking it to a rise in temperature.

Section II – Options

Question 28 – Communication

- (a) (i) Better responses were concise and did not attempt to give additional information. Weaker responses often showed confusion over the term ‘refractive media’.
- (ii) Better responses clearly showed how a problem with one of the refractive media could cause defective vision. Weaker responses simply described vision problems such as cataracts or myopia or demonstrated a limited knowledge of eye anatomy.
- (b) (i) Better responses provided the main characteristics and features of an appropriate first-hand investigation. These responses included a clear link to the secondary sources used to locate the regions involved in speech, sight and sound perception. In many weaker responses, candidates provided irrelevant information about the structure of the brain.
- (ii) Better responses related relevance to the investigation and reliability to the process of gathering information. Weaker responses did not distinguish between the terms *relevance* and *reliability*.

- (c) In better responses, candidates differentiated between the structures that produce, detect and perceive sound, detailing how they work to enable effective communication. Their responses demonstrated coherence and the correct use of scientific principles. Weaker responses often confused detection and perception and identified the structures without making a judgement about their effectiveness in communication.
- (d) (i) Better responses selected the relevant graph and identified a wavelength between 530nm and 550nm for peak absorption.
- (ii) Better responses used quantitative data to describe the relevant graph. These responses indicated the different functions of each type of pigment cone and provided a reason for the necessity of all three in colour vision.
- (iii) In better responses, candidates made the relationship between the occurrence of colour vision and its use for communication evident.

Question 29 – Biotechnology

- (a) (i) In better responses, candidates named organic compounds produced by fermentation since the early eighteenth century. Weaker responses confused earlier forms of fermentation naming compounds such as alcohol and bread.
- (ii) Better responses clearly explained the use of one organic compound and the impact it had on society at the time of its introduction.
- (b) (i) In better responses, candidates provided features and characteristics of ways in which the information on a current application of transgenic technology was gathered and analysed. Weaker responses described gathering and analysing information in general terms without linking it to the investigation.
- (ii) Better responses related relevance to the investigation and reliability to the process of gathering information. Weaker responses did not distinguish between the terms *relevance* and *reliability*.
- (c) In better responses, candidates outlined biotechnological processes used in aquaculture and provided detailed examples. The judgement provided in the response was coherent and included the correct use of scientific principles. Weaker responses identified some examples of aquaculture but not the processes involved. Weaker responses were often excessively long, showing little evidence of planning or structure.
- (d) (i) Better responses estimated the time at which ethanol concentration was at a maximum by using a narrow range or single number value.
- (iii) In better responses, candidates explained how at least two changes in technology have modified traditional uses of biotechnology. Weaker responses mentioned the changes without the link to traditional processes.

Question 30 – Genetics: The Code Broken?

- (a) (i) In better responses, candidates correctly identified two examples of characteristics determined by multiple alleles in an organism other than a human.
- (ii) Better responses clearly showed how one of these characteristics provided variability in the phenotype. Weaker responses simply described the variability in the phenotype.
- (b) (i) In better responses, candidates were explicit in providing features and characteristics of how the information on the cloning of an animal was gathered. In addition, these responses clearly stated how the information was analysed. Weaker responses were general and did not indicate a link to animal cloning.
- (ii) Better responses related relevance to the investigation and reliability to the process of gathering information.
- (c) In better responses, candidates outlined the techniques of mapping and gene therapy to manage a genetic disease such as cystic fibrosis, giving a strong judgement for both techniques. Weaker responses addressed the Human Genome Project rather than the mapping aspect and judgement statements were often general or absent.
- (d) (ii) Better responses defined *trisomy* as an extra chromosome of an existing homologous pair of chromosomes.
- (iii) Better responses described the mechanism of gene relocation and provided points to illustrate how these transposals increased genetic diversity or variability.

Question 31 – The Human Story

- (a) (i) Weaker responses incorrectly named their examples with other applied names such as 'Lucy' or 'the Hobbit'.
- (ii) Better responses clearly compared features of their named hominid with *Homo sapiens*. These answers were often tabulated. Weaker responses made incorrect comparisons or simply listed features of the hominid or *Homo sapiens*.
- (b) (i) Better responses described how information was processed and gathered and how it was assessed. Weaker responses indicated confusion with gathering and processing information.
- (ii) Better responses provided features of reliable data collection and identified how to judge the relevance of the material.
- (c) In better responses, candidates made a clear judgement about the impact of cultural development throughout human evolution. The judgement was demonstrated in a logical and coherent response. In weaker responses, candidates listed cultural developments without reference to their impact. Weaker responses were excessively long, showing little evidence of planning and structure.
- (d) (ii) In better responses, candidates used the graph to provide features of a change in the human population and suggested a specific reason for the change. Weaker responses suggested a reason that did not relate to the change.

- (iii) Better responses named a modern technology such as gene therapy or IVF, identified issues associated with the potential impact of the technology and provided points for and/or against the impact of the technology. Weaker responses did not name a modern technology.

Question 32 – Biochemistry

- (a)
 - (i) Better responses correctly identified products of photosynthesis such as glucose and oxygen.
 - (ii) Better responses identified the nature of a product named in (a) (i) and showed how it was used in plants. Weaker responses did not refer to the biochemical processes and used general non-specific explanations like ‘used to keep cells alive’ or ‘used to feed the plant’.
- (b)
 - (i) Better responses described how the data on chloroplast size, shape and distribution was collected and then how this data was presented to show a comparison between angiosperms.
 - (ii) Better responses provided features of reliable data collection and identified how to judge the relevance of the material. These judgements were made in the context of gathering and presenting information about the shape, size and distribution of chloroplasts in angiosperms.
- (c) In better responses, candidates correctly named at least two isotopes and outlined the role of each in photosynthetic research. These responses described examples of research where the isotopes had been used and gave a judgement of how this research has improved our understanding of photosynthetic pathways. Weaker responses did not name the isotopes or did not give specific information of their use in photosynthetic research.
- (d)
 - (ii) Better responses identified the thylakoids as the location for chlorophyll *a* absorption and described the absorption trends using specific reference to wavelengths.
 - (iii) Better responses identified at least two other pigments (eg chlorophyll *b*, carotenoids, xanthophylls). These responses provided information about the role of these pigments in the photosynthetic process. Many weaker responses identified the role of pigments in terms of the colour of the leaves of plants.

Biology

2007 HSC Examination Mapping Grid

Question	Marks	Content	Syllabus outcomes
Section I			
Part A			
1	1	9.2.1.2.1, 12.3(c)	H6, H12
2	1	9.2.1.2.1	H6
3	1	9.2.2.2.6, 9.2.2.3.6, 12.3(c)	H6, H12
4	1	9.4.7.2.1, 9.4.7.3.2, 14.3(c)	H8, H14
5	1	9.4.2.3.2, 11.2(a)	H11
6	1	9.4.4.2.4	H6
7	1	9.2.3.2.9	H7
8	1	9.4.3.2.1	H6
9	1	9.3.2.2.1, 9.3.2.2.2, 9.3.5.2.1	H9
10	1	9.3.4.2.3	H6, H9
11	1	9.3.2.2.3, 9.3.2.2.4, 9.3.2.3.1 12.3(c), 14.1(a)	H9, H12, H14
12	1	9.4.4.2.2, 9.4.5.2.1, 9.4.5.2.2, H14.1(a)	H6, H14
13	1	9.3.1.2.3, 9.3.1.3.2, 14.3(c)	H10, H14
14	1	9.2.3.2.7	H6
15	1	9.4.3.2.1	H9
Section I			
Part B			
16(a)	2	9.4.2.2.2, 14.3(d)	H8, H14
16(b)	1	9.4.2.2.2	H8
17	3	9.2.3.3.4, 13.1(e)	H6, H13
18 (a)	2	9.2.2.3.4	H6
18 (b)	2	9.2.2.3.5	H5
19 (a)	1	9.4.3.2.2	H6
19 (b)	2	9.4.3.2.2	H6

Question	Marks	Content	Syllabus outcomes
19 (c)	3	9.4.4.2.1, 9.4.3.3.3	H6
20 (a)	3	9.3.3.2.3	H9
20 (b)	3	9.3.4.2.1, 13.1(e)	H9, H13
21 (a)	2	9.4.3.3.3, 9.4.3.2.2, 12.3(c), 14.1(a, g)	H12, H14
21 (b)	4	9.4.5.3.1	H4, H6
22 (a)	2	9.3.3.2.4, 9.3.3.3.1, 12.3(c)	H9, H12
22(b)	2	9.3.3.2.4, 9.3.3.3.1, 13.1(e)	H9, H13
22(c)	2	9.3.3.2.5, 9.3.4.2.6	H9
22(d)	2	9.3.2.2.5	H9
23	4	9.1, 11.2 (b, c), 11.3 (a)	H11
24 (a)	1	9.2.2.2.5	H6
24 (b)	2	9.2.2.3.3, 12.3(c)	H3, H12
24 (c)	3	9.2.2.4, 9.2.2.5	H6
25	3	9.4.6.3.1, 9.4.6.2.1, 14.1 (a, c, g)	H14
26	3	9.4.1.2.2, 9.4.4, 9.4.5, 9.4.6	H6
27	8	9.2.3.2.9, 9.3.4.3.4, 9.3.1.2.1, 9.3.4.2.6, 9.2.1.2.7.8.9, 9.2.1.3.3, 14.3(b)	H4, H7, H8, H10, H14
Section II Communication			
28 (a) (i)	2	9.5.3.2.2	H6
28 (a) (ii)	3	9.5.3.2.2, 9.5.3.3.3, 9.5.3.2.5	H6
28 (b) (i)	3	9.5.7.3.2, 11.2(d), 12.2(b), 12.4 (a, c, d)	H6, H11, H12
28 (b) (ii)	3	9.1, 12.4 (d, e)	H12
28 (c)	7	9.5.1.2.1, 9.5.5.2.1, 9.5.5.2.3, 9.5.6.2.2, 9.5.6.2.5, 9.5.7.2.4, 9.5.6.3.1, 14.3(b)	H6, H14
28 (d) (i)	1	12.3 (e)	H12
28 (d) (ii)	2	9.5.4.2.4, 14.1(a)	H14
28 (d) (iii)	4	9.5.4.3.2, 14.1(h)	H6, H14

Question	Marks	Content	Syllabus outcomes
Section II			
Question 29— Biotechnology			
29 (a) (i)	2	9.6.3.2.1	H8
29 (a) (ii)	3	9.6.3.3.1	H8
29 (b) (i)	3	9.6.5.3.3, 12.3 (b, c, d), 14.1(a, c)	H6, H12, H14
29 (b) (ii)	3	12.4 (d, e)	H12
29 (c)	7	9.6.6.3.1, 9.6.7.3.1, 9.6.6.2.4, 14.3(b)	H3, H14
29 (d) (i)	1	9.1.12.3, 12.3(c)	H6, H12
29 (d) (ii)	2	9.6.2.2.1, 14.1(a)	H6, H14
29 (d) (iii)	4	9.6.3.3.2	H2, H6
Section II			
Genetics: The Code Broken?			
30 (a) (i)	2	9.7.2.2.1	H9
30 (a) (ii)	3	9.7.2.2.3, 9.7.2.3.2, 9.7.2.1, 9.7.2.2.1	H9
30 (b) (i)	3	9.7.7.3.2, 12.3 (a, b, c, d), 14.1 (a, e)	H9, H12, H14
30 (b) (ii)	3	12.4 (d, e)	H12
30 (c)	7	9.7.4, 9.7.5, 9.7.6.3.1, 14.3(b)	H4, H14
30 (d) (i)	1	12.3(e)	H9, H12
30 (d) (ii)	2	9.7.6.2.1, 14.1(a)	H9, H14
30 (d) (iii)	4	9.7.6.2.3	H9
Section II			
The Human Story			
31 (a) (i)	2	9.8.3.2.1	H10
31 (a) (ii)	3	9.8.3.2.2	H10
31 (b) (i)	3	9.8.3.3.2, 12.4 (a, c), 14.3 (c, d)	H10, H12, H14
31 (b) (ii)	3	9.8.2.2.5, 12.4 (d, e)	H12
31 (c)	7	9.8.3.2.2, 9.8.5, 9.8.6, 14.3(b)	H10, H14
31 (d) (i)	1	12.3(c)	H12
31 (d) (ii)	2	9.8.6.3.1, 14.1(a)	H10, H14

Question	Marks	Content	Syllabus outcomes
31 (d) (iii)	4	9.8.6.3.1, 14.3(c)	H10, H14
Section II Biochemistry			
32 (a) (i)	2	9.9.1.3.1	H6
32 (a) (ii)	3	9.9.1.3.1	H6
32 (b) (i)	3	9.9.8.3.1, 12.3 (a, b, c), 13.1 (a, b, c, e)	H6, H12
32 (b) (ii)	3	12.4 (d, e)	H6, H12
32 (c)	7	9.9.6.2.1, 9.9.5.1.3, 9.9.9.3.2, 14.3(b)	H14
32 (d) (i)	1	12.3 (c)	H12
32 (d) (ii)	2	9.9.8.3.2, 9.9.4.2.4, 14.1(a)	H6, H14
32 (d) (iii)	4	9.9.3.2.5, 9.9.3.3.4	H6

2007 HSC Biology Marking Guidelines

Section I, Part B

Question 16 (a)

Outcomes assessed: H8, H14

MARKING GUIDELINES

Criteria	Marks
Identifies a source of pathogens and relates the wearing of hat and/or gloves as a type of barrier to pathogen transfer	2
- Identifies a source of pathogens OR - Identifies a barrier to pathogen transfer	1

Question 16 (b)

Outcomes assessed: H8

MARKING GUIDELINES

Criteria	Marks
Names a hygiene practice	1

Question 17

Outcomes assessed: H6, H13

MARKING GUIDELINES

Criteria	Marks
- Provides a table that shows difference of marine and freshwater fish in terms of: – environmental conditions – urine production	3
- Provides a table that shows difference of marine and freshwater fish in terms of either: – environmental conditions or urine production OR - Shows differences without the use of a table	2
- Constructs a table with appropriate headings (columns and rows) OR - Identifies a difference without the use of a table	1

Question 18 (a)

Outcomes assessed: H6

MARKING GUIDELINES

Criteria	Marks
Sketches in general terms how a named extracted blood product is used to restore normal body function	2
- Identifies an extracted blood product OR - Identifies a use of a product extracted from donated blood	1

Question 18 (b)

Outcomes assessed: H5

MARKING GUIDELINES

Criteria	Marks
Puts forward relevant reasons	2
Puts forward a relevant reason	1

Question 19 (a)

Outcomes assessed: H6

MARKING GUIDELINES

Criteria	Marks
Correctly names one disease caused by a macro-parasite	1

Question 19 (b)

Outcomes assessed: H6

MARKING GUIDELINES

Criteria	Marks
• Correctly identifies distinguishing features of prions from protozoans	2
• Correctly identifies a distinguishing feature of prions from protozoans	1

Question 19 (c)

Outcomes assessed: H6

MARKING GUIDELINES

Criteria	Marks
• For a named infectious disease identifies: – the mode of transmission – the entry point – an immune response	3
• For a named infectious disease identifies TWO of: – the mode of transmission – the entry point – an immune response	2
• Identifies a mode of transmission or the entry point or an immune response of an infectious disease	1

Question 20 (a)

Outcomes assessed: H9

MARKING GUIDELINES

Criteria	Marks
• Completes each row correctly	3
• Completes TWO rows correctly OR • Completes base column correctly • Provides correct % for T OR C/G	2
• Completes ONE row correctly OR • Completes base column correctly	1

Question 20 (b)*Outcomes assessed: H9, H13***MARKING GUIDELINES**

Criteria	Marks
• Provides a simple flowchart that provides features and characteristics of DNA replication in the correct sequence	3
• Provides a simple flowchart “or numbered sequence” that outlines DNA replication in the correct sequence	2
• Identifies the stages of DNA replication in the correct sequence with at least one step	1

Question 21 (a)*Outcomes assessed: H12, H14***MARKING GUIDELINES**

Criteria	Marks
• Identifies the most dangerous disease and provides two valid reasons	2
• Provides a valid reason OR • Identifies the most dangerous disease	1

Question 21 (b)

Outcomes assessed: H4, H6

MARKING GUIDELINES

Criteria	Marks
- Identifies a disease from the table - Describes the vaccination program for this disease - Makes a judgment about the effectiveness of the program	3-4
- Describes features of a vaccination program or vaccination for a named disease. OR - Describes a feature of a vaccination for a named disease and makes a judgement about the effectiveness of the program	2
- Outlines one feature of a vaccination for a named disease OR - Make a judgement for a named disease without reference to the program	1

Question 22 (a)

Outcomes assessed: H9, H12

MARKING GUIDELINES

Criteria	Marks
Provides the features of the behaviour of chromosomes between the steps by indicating the pairing of homologous chromosomes and crossing over	2
Identifies a feature of the behaviour of chromosomes between the steps	1

Outcomes assessed: H9, H13

MARKING GUIDELINES

Criteria	Marks
Provides four possible combinations	2
Provides two or three possible combinations	1

Question 22 (c)

Outcomes assessed: H9

MARKING GUIDELINES

Criteria	Marks
Shows how one result of the process is an advantage for a species	2
Identifies either a result of the process or an advantage of the process	1

Question 22 (d)

Outcomes assessed: H9

MARKING GUIDELINES

Criteria	Marks
• Provides the difference between the terms allele and gene	2
• Provides an outline of either an allele OR a gene	1

Question 23

Outcomes assessed: H11

MARKING GUIDELINES

Criteria	Marks
• Provides an ordered relevant procedure that: – states variables to be controlled – describes ways to collect reliable and valid data information – identifies equipment needed to undertake the investigation	4
• Provides an ordered relevant procedure • Provides details of at least TWO components of a controlled experiment	3
• Provides a relevant procedure • Provides details of ONE component of a controlled experiment	2
• Provides a procedure outlined in general terms only	1

Question 24 (a)

Outcomes assessed: H6

MARKING GUIDELINES

Criteria	Marks
• Provides a correct reason for monitoring/need for oxygen	1

Question 24 (b)

Outcomes assessed: H3, H12

MARKING GUIDELINES

Criteria	Marks
• Identifies one improvement in the T-Stat oximeter over the pulse oximeter • Gives a reason why this is advantageous	2
• Identifies one improvement in the T-Stat oximeter over the pulse oximeter	1

Question 24 (c)

Outcomes assessed: H6

MARKING GUIDELINES

Criteria	Marks
- Provides features of changes in the chemical composition of blood as it moves along a capillary - States reasons for the changes	3
- Provides features of changes in the chemical composition of blood as it moves along a capillary OR - Provides a feature of a chemical change and a reason for the change	2
Provides a feature of a chemical change	1

Question 25

Outcomes assessed: H14

MARKING GUIDELINES

Criteria	Marks
- Identifies characteristics of a sound epidemiological study - Links the type of data that needed to be collected to establish the relationship between smoking and lung cancer	3
- Identifies characteristics of a sound epidemiological study - Identifies one type of data that needed to be collected to establish the relationship OR - Identifies the types of data that needed to be collected to establish the relationship	2
- Identifies a characteristic of an epidemiological study OR - Identifies one type of data that needed to be collected to establish the relationship	1

Question 26

Outcomes assessed: H6

MARKING GUIDELINES

Criteria	Marks
• Provides features of how genes assist in the maintenance of health	3
• Outlines how genes assist in the maintenance of health	2
• Provides an example OR • States a role of genes	1

Question 27

Outcomes assessed: H4, H7, H8, H10, H14

MARKING GUIDELINES

Criteria	Marks
- Explains possible consequences of temperature change for plants and animals using examples of individuals and species - Relates implications of changes in survival of individuals to survival of species - Demonstrates coherence and logical progression and includes correct use of scientific principles and ideas	7-8
- Explains possible consequences of temperature change for plants and/or animals using examples that may have limitations - Describes some possible implications for individuals and species	5-6
- Describes some plant or animal responses to temperature change - Provides a statement about survival of individuals or extinction of species	3-4
• States some responses of plants and/or animals to temperature change.	1-2

Section II

Question 28 (a) (i)

Outcomes assessed: H6

MARKING GUIDELINES

Criteria	Marks
• Identifies TWO refractive media in the human eye	2
• Identifies ONE refractive medium in the human eye	1

Question 28 (a) (ii)

Outcomes assessed: H6

MARKING GUIDELINES

Criteria	Marks
• Shows how a problem may contribute to poor eye sight or blindness	3
• Describes the problem/change in the medium OR • Describes the cause or outcome of the problem	2
• States one possible change in one refractive medium	1

Question 28 (b) (i)

Outcomes assessed: H6, H11, H12

MARKING GUIDELINES

Criteria	Marks
• Provides features and characteristics of how to undertake the investigation showing a link to the way the information was gathered	3
• Outlines how to undertake the investigation showing a link to the way the information was gathered OR • Provides features and characteristics of how to undertake the investigation or how the information was gathered	2
• Provides some steps to undertake the investigation	1

Question 28 (b) (ii)

Outcomes assessed: H12

MARKING GUIDELINES

Criteria	Marks
States how both relevance and reliability of information can be judged and is linked to this investigation	3
- States how both relevance and reliability of information can be judged OR - States how relevance or reliability can be judged and how it relates to this investigation	2
States how either relevance or reliability could be judged	1

Question 28 (c)

Outcomes assessed: H6, H14

MARKING GUIDELINES

Criteria	Marks
- Names structures that produce, detect and perceive sounds - Outlines the roles of these structures - Makes a judgement about the value of each of these in effective human communication - Demonstrates coherence and logical progression and includes correct use of scientific principles and ideas.	6-7
- Names structures that are involved in communication using sound - States the role of some of the structures - Makes a judgement about the role of one of these in effective communication	4-5
- Names some relevant structures - States the role of some of these structures in effective communication	1-3

Question 28 (d) (i)

Outcomes assessed: H12

MARKING GUIDELINES

Criteria	Marks
Estimates wavelength the green pigment cone absorbs the most light.	1

Question 28 (d) (ii)

Outcomes assessed: H14

MARKING GUIDELINES

Criteria	Marks
- Provides features of the pattern of light absorption for red pigment cones - States a correct reason for the need for 3 different pigments for colour vision	2
Any one of the above	1

Question 28 (d) (iii)

Outcomes assessed: H6, H14

MARKING GUIDELINES

Criteria	Marks
- Identifies the use of colour for communication in animals - Identifies the occurrence of colour vision in animals - Makes the relationship between the use of colour and the occurrence of colour vision evident	4
- Identifies the use of colour for communication in animals - Identifies the occurrence of colour vision in animals	2-3
- Identifies the use of colour for communication in animals OR - Identifies the occurrence of colour vision in animals	1

Question 29 (a) (i)

Outcomes assessed: H8

MARKING GUIDELINES

Criteria	Marks
Identifies organic compounds produced by fermentation techniques	2
Identifies an organic compound produced by fermentation techniques	1

Question 29 (a) (ii)

Outcomes assessed: H8

MARKING GUIDELINES

Criteria	Marks
Shows how the use of an organic compound had an impact on society at the time of its introduction.	3
- Describes the use of an organic compound OR - Describes the impact of the compound on society at the time of its introduction	2
- States a use of the compound OR - States an impact on society	1

Question 29 (b) (i)

Outcomes assessed: H6, H12, H14

MARKING GUIDELINES

Criteria	Marks
Provides features and characteristics of how the information was gathered and analysed in this investigation	3
- Outlines how the information was gathered and analysed in this investigation OR - Provides features and characteristics of how the information was gathered or analysed in this investigation	2
Provides some steps to gather or analyse the information	1

Question 29 (b) (ii)

Outcomes assessed: H12

MARKING GUIDELINES

Criteria	Marks
States how both relevance and reliability of information can be judged and is linked to this investigation	3
- States how both relevance and reliability of information can be judged OR - States how relevance or reliability can be judged and how it relates to this investigation	2
States how either relevance or reliability could be judged	1

Question 29 (c)

Outcomes assessed: H3, H14

MARKING GUIDELINES

Criteria	Marks
- Names techniques of aquaculture - Outlines how these techniques are used - Makes a judgement about the value of the techniques - Demonstrates coherence and logical progression and includes correct use of scientific principles and ideas.	6-7
- Names techniques of aquaculture - States the uses of the techniques - Makes a judgement about the role of ONE of these techniques	4-5
- Names a relevant technique of aquaculture - States the use of the technique	2-3
Names a relevant technique of aquaculture	1

Question 29 (d) (i)

Outcomes assessed: H12

MARKING GUIDELINES

Criteria	Marks
Estimates a time at which ethanol concentration is at its maximum.	1

Question 29 (d) (ii)

Outcomes assessed: H6, H14

MARKING GUIDELINES

Criteria	Marks
- Provides features of the trend in glucose concentration - States a correct reason for this pattern	2
Any ONE of the above	1

Question 29 (d) (iii)

Outcomes assessed: H2, H6

MARKING GUIDELINES

Criteria	Marks
- Identifies traditional uses of biotechnology - Shows how changes in technology have modified traditional uses	4
- Identifies traditional uses of biotechnology - Identifies changes in the technology that have modified traditional uses	3
- Identifies a traditional use of biotechnology - Identifies a technological change in this traditional use	2
- Identifies a traditional use of biotechnology OR - Names an appropriate example	1

Question 30 (a) (i)

Outcomes assessed: H9

MARKING GUIDELINES

Criteria	Marks
Identifies characteristics determined by multiple alleles in a non-human organism	2
Identifies a characteristics determined by multiple alleles in a non-human organism	1

Question 30 (a) (ii)

Outcomes assessed: H9

MARKING GUIDELINES

Criteria	Marks
Shows how one of these characteristics provide variability in phenotypes	3
- Describes variation in the phenotype OR - Describes variation in the genotype	2
States a variation in the phenotype or genotype	1

Question 30 (b) (i)

Outcomes assessed: H9, H12, H14

MARKING GUIDELINES

Criteria	Marks
Provides features and characteristics of how the information was gathered and analysed in this investigation.	3
- Outlines how the information was gathered and analysed in this investigation OR - Provides features and characteristics of how the information was gathered or analysed in this investigation	2
Provides some steps to gather or analyse the information	1

Question 30 (b) (ii)

Outcomes assessed: H9, H12

MARKING GUIDELINES

Criteria	Marks
States how both relevance and reliability of information can be judged and is linked to this investigation	3
- States how both relevance and reliability of information can be judged OR - States how relevance or reliability can be judged and how it relates to this investigation	2
States how either relevance or reliability could be judged	1

Question 30 (c)

Outcomes assessed: H4, H14

MARKING GUIDELINES

Criteria	Marks
• Outlines in general terms the techniques • Gives an example of each • Makes a judgment about the value of each of the techniques in managing a genetic disease or a form of cancer or AIDS • Demonstrates coherence and logical progression and includes correct use of scientific principles and ideas.	6-7
• Outlines in general terms the techniques • Gives an example • Makes a statement about the value of the technique	4-5
• Outlines in general terms a technique • Gives an example	2-3
• Outlines an appropriate technique	1

Question 30 (d) (i)

Outcomes assessed: H9, H12

MARKING GUIDELINES

Criteria	Marks
• Estimates a number of infants with Down syndrome per 1000 births for 40-year-old mothers	1

Question 30 (d) (ii)

Outcomes assessed H9, H14

MARKING GUIDELINES

Criteria	Marks
• Provides features of the trend • Provides a correct definition of the term <i>trisomy</i>	2
• Provides one of the above	1

Question 30 (d) (iii)

Outcomes assessed: H9

MARKING GUIDELINES

Criteria	Marks
- Defines transposable genetic elements - Provides points for and/or against how these transposable genetic elements increase genetic diversity or variability - Uses example(s) to support discussion	4
- Defines transposable genetic elements - Provides points for and/or against how these transposable genetic elements increase generic diversity or variability	3
- Defines transposable genetic elements - States how these elements increase genetic diversity or variability	2
Defines transposable genetic elements	1

Question 31 (a) (i)

Outcomes assessed: H10

MARKING GUIDELINES

Criteria	Marks
Identifies TWO hominids known from fossil evidence	2
Identifies ONE hominid known from fossil evidence	1

Question 31 (a) (ii)

Outcomes assessed: H10

MARKING GUIDELINES

Criteria	Marks
Shows how features and characteristics of one hominid (from a i) are different from <i>Homo sapiens</i>	3
- Describes features of one hominid - Describes features of <i>Homo sapiens</i> but does not show how the listed features are different	2
- States a difference between one hominid and <i>Homo sapiens</i> OR - Outlines features of either a hominid or <i>Homo sapiens</i>	1

Question 31 (b) (i)

Outcomes assessed: H10, H12

MARKING GUIDELINES

Criteria	Marks
Provides features and characteristics of how the information was processed and how available evidence was used in the investigation.	3
- Outlines how the information was processed and how available evidence was used in the investigation OR - Provides features and characteristics of how the information was processed OR available evidence used	2
Provides some steps to undertake the investigation	1

Question 31 (b) (ii)

Outcomes assessed: H12

MARKING GUIDELINES

Criteria	Marks
States how both relevance and reliability of information can be judged and is linked to this investigation	3
- States how both relevance and reliability of information can be judged OR - States how relevance or reliability can be judged and how it relates to this investigation	2
States how either relevance or reliability could be judged	1

Question 31 (c)

Outcomes assessed: H10, H14

MARKING GUIDELINES

Criteria	Marks
- Names examples of cultural development - Outlines how EACH of these has impacted - Makes a judgement about the impact of cultural development - Demonstrates coherence and logical progression and includes correct use of scientific principles and ideas	6–7
- Names examples of cultural development - Outlines how EACH of these has impacted - Makes a judgement about the impact of ONE of these examples	4–5
Names an example of cultural development AND outlines its impact	2–3
Names an example of cultural development	1

Question 31 (d) (i)

Outcomes assessed: H12

MARKING GUIDELINES

Criteria	Marks
Estimates the human population 1000 years ago	1

Question 31 (d) (ii)

Outcomes assessed: H10, H14

MARKING GUIDELINES

Criteria	Marks
- Provides features of a change - Suggests a reason for this change	2
- Identifies a change - Suggests a reason for this change	1

Question 31 (d) (iii)

Outcomes assessed: H14

MARKING GUIDELINES

Criteria	Marks
- Names a technology which could impact on future human populations - Identifies issue(s) relating to the potential impact of the technology - Provides point(s) for and/or against the impact of the technology	3–4
- Names a technology which could impact on future human populations AND/OR - States a potential impact of the technology	1–2

Question 32 (a) (i)

Outcomes assessed: H6

MARKING GUIDELINES

Criteria	Marks
Identifies TWO products of photosynthesis	2
Identifies ONE product of photosynthesis	1

Question 32 (a) (ii)

Outcomes assessed: H6

MARKING GUIDELINES

Criteria	Marks
Shows a relationship between the product and its use in living organisms	3
- Describes a product OR - Describes the use of a product in living organisms	2
States the use of a product in an organism	1

Question 32 (b) (i)

Outcomes assessed: H6, H12, H13

MARKING GUIDELINES

Criteria	Marks
Provides features and characteristics of how the information was gathered and presented in the investigation	3
- Outlines how the information was gathered and presented in the investigation OR - Provides features and characteristics of how the information was gathered or presented in the investigation	2
Provides some steps to undertake the investigation	1

Question 32 (b) (ii)

Outcomes assessed: H12

MARKING GUIDELINES

Criteria	Marks
States how both relevance and reliability of information can be judged and is linked to this investigation	3
- States how both relevance and reliability of information can be judged OR - States how relevance or reliability can be judged and how it relates to this investigation	2
States how either relevance or reliability could be judged	1

Question 32 (c)

Outcomes assessed: H1, H2, H5

MARKING GUIDELINES

Criteria	Marks
- Names isotopes - Outlines the role of EACH in developing our understanding of photosynthesis - Makes a judgement about the use of isotopes in relation to photosynthesis - Demonstrates coherence and logical progression and includes correct use of scientific principles and ideas.	6–7
- Names isotopes - Outlines the role of EACH in developing our understanding of photosynthesis	4–5
- Names an isotope AND - Outlines its role in developing our understanding of photosynthesis	2–3
- Names an isotope OR - Makes a general statement about the use of isotopes in developing our understanding of photosynthesis.	1

Question 32 (d) (i)

Outcomes assessed: H12

MARKING GUIDELINES

Criteria	Marks
Correctly estimates the wavelength at which most photosynthesis occurs	1

Question 32 (d) (ii)

Outcomes assessed: H14

MARKING GUIDELINES

Criteria	Marks
- States where absorption occurs within a chloroplast - Provides features of the absorption of chlorophyll <i>a</i> over the visible light spectrum	2
- States where absorption occurs within a chloroplast OR - Provides features of absorption of chlorophyll <i>a</i> over the visible light spectrum	1

Question 32 (d) (iii)*Outcomes assessed: H5***MARKING GUIDELINES**

Criteria	Marks
- Names other pigments - Provides point(s) for and/or against the presence of these pigments	3–4
- Names another pigment AND/OR - States the role of this pigment in photosynthesis	1–2